

PhysGest: Transforming Static Textbook Diagrams into Physically Realistic Videos for Enhanced Physics Education

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Abstract

Key physics concepts and principles, such as momentum conservation, are fundamental to the mechanics section of physics education. They play a crucial role in understanding the interactions between objects and the motion laws governing systems. However, since these processes occur instantaneously and are difficult to observe, traditional teaching methods, such as chalkboard explanations, static textbook diagrams, or apparatus demonstrations, often fail to effectively convey the dynamic changes involved. Additionally, these traditional methods cannot effectively demonstrate how these processes comply with physical laws, such as the conservation of momentum. As a result, students struggle to grasp the underlying principles. To address these issues, this paper proposes PhysGest, a framework that transforms static textbook diagrams into physically accurate and interactive videos for physics education. PhysGest captures the instructor's gestures and extracts temporal motion trajectories, from which physical parameters such as velocity and direction are derived. These extracted parameters are then used to animate the static diagrams in textbooks, generating dynamic videos that align with the instructor's gestures and adhere to physical laws. Therefore, PhysGest overcomes the lack of interaction by converting static diagrams into videos that enhance students' ability to understand physical concepts. Our evaluation results demonstrate that PhysGest significantly enhances students' self-efficacy, perceived enjoyment, and knowledge retention with improved performance in understanding underlying concepts and principles.

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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Modeling and simulation*; • **Applied computing** → *Education*.

ACM Reference Format:

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1 Introduction

Some core physics concepts like momentum conservation are fundamental yet challenging to teach, as their instantaneous nature is difficult to convey with traditional teaching with chalkboards or static diagrams in textbooks. With the rapid development of Artificial Intelligence (AI) and video generation technologies, particularly the integration of Image-to-Video (I2V) generation, new opportunities have emerged for advancing traditional physics education [Wei et al. 2024]. Existing studies have explored the use of generative models to produce dynamic animations from static images, thereby enhancing the visual representation of physical processes [Ni et al. 2023]. However, most of these models prioritize image coherence and visual realism while neglecting adherence to fundamental physical laws such as momentum conservation and energy transformation, and collision types [Gunturu et al. 2024]. This often leads to distorted motion logic or even results that contradict real-world physics. Moreover, current educational tools lack sufficient interactivity: students are unable to actively regulate the process, and essential physical quantities, such as changes in velocity or momentum trends, are not effectively visualized. As a result, physics education still lacks an operational, reliable, and accessible mechanism that ensures physical consistency while providing students with an engaging, participatory deep learning experience.

In the mechanics section of physics education, the law of conservation of momentum and collisions constitute core knowledge points for understanding the interactions between objects and the rules governing their motion. However, traditional textbooks typically present this content through abstract formulas and static

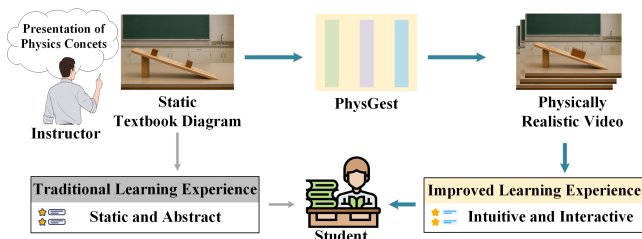


Figure 1: The left shows the traditional teaching approach, relying on static textbook diagrams to explain physics concepts (e.g., momentum conservation), which is static and abstract. In contrast, the right shows our PhysGest framework, generating physically realistic and interactive video from a static textbook diagram, offering a more intuitive and interactive learning experience.

diagrams, lacking dynamic reconstructions of physical processes [Gunturu et al. 2024]. As a result, students often struggle to establish cognitive connections between formulas and real-world phenomena [Saifullah et al. 2017]. Such abstraction imposes high demands on students’ logical reasoning and spatial construction abilities, and misconceptions frequently arise when analyzing momentum transfer and conservation processes [Sakti et al. 2024]. For example, collision experiments in classrooms are often constrained by short time frames, invisible variables, and coarse experimental outcomes, making it difficult for students to intuitively grasp collision mechanisms. Even with demonstration apparatus, limitations such as uncontrollable parameters, poor reproducibility, and high costs further reduce the effectiveness of experimental teaching. Overall, current instruction on some key physical principles, such as momentum conservation, still faces multiple challenges, including strong abstraction, invisibility of processes, and limited experimental control, highlighting the need for more interactive and visualized teaching support methods.

To address the aforementioned issues, this study proposes PhysGest as illustrated in Figure 1. PhysGest is a generative teaching framework tailored to education on specific physics concepts and principles, such as collisions and the law of conservation of momentum, which are difficult to observe as they occur instantaneously. PhysGest integrates diagram parsing and physical simulation to transform static textbook diagrams into dynamic demonstrations that can also respond to gestures from the user (instructor or potentially from the student). By assigning initial velocities to objects through hand gestures, the static diagrams are animated to align with the user’s gestures, adhering to physical laws. For example, PhysGest can generate object collision videos through its dynamic physical simulations that are consistent with the principle of momentum conservation. Our evaluation results show that PhysGest significantly enhances students’ self-efficacy, perceived enjoyment, and knowledge retention, with improved performance in understanding underlying concepts and principles and in conducting

physics reasoning. These results demonstrate the strong pedagogical applicability of PhysGest and its potential for widespread adoption.

2 Related Work

2.1 Image-to-Video Generation

In recent years, Image-to-Video (I2V) generation has made significant progress with the advancement of diffusion models [Ho et al. 2020], enabling the synthesis of visually coherent short videos with applications in character animation and virtual reality [Karras et al. 2023]. For example, DreamVideo [Wei et al. 2024] achieves motion generation by preserving image details and following textual guidance, employing a frame-preservation branch based on pre-trained video diffusion models for high-fidelity synthesis. “Animate Anyone” [Hu 2024] integrates image details with pose guidance through ReferenceNet to ensure consistency in character animation. ControlNext [Peng et al. 2024] achieves efficient image and video generation through architectural simplification and cross-normalization. MOFA-Video [Niu et al. 2024] generates motion fields adaptable to multiple control signals, improving the controllability of motion trajectories. Furthermore, Lestari et al. [Lestari et al. 2025] highlighted the potential of AI-generated videos in physics education, offering accurate motion representations that enhance immersive learning. Kumar et al. [Kumar et al. 2024] explored AI’s transformative role in dynamically adjusting educational content, while Netland et al. [Netland et al. 2025], in an experimental study, found that AI-generated teaching videos can lead to learning outcomes comparable to those from human-made videos. However, these existing approaches emphasize visual coherence and realism while neglecting adherence to fundamental physical laws. As a result, current educational practice still lacks an operable, reliable, and low-threshold auxiliary mechanism that can ensure physical consistency while offering students an interactive and participatory deep learning experience.

2.2 Digital Demonstration Tools in Physics Education

Educational information technologies are driving physics teaching toward more intuitive perception, with digital tools leveraging interactive simulations to enhance students’ understanding of momentum conservation. In this field, PhET [Perkins et al. 2006] offers interactive collision simulations with adjustable parameters and real-time data charts, but its predefined scenarios often fail to align with textbook content, limiting exploratory experimentation. Algodoo [Gregorcic and Bodin 2017], a 2D sandbox, allows users to adjust parameters such as friction coefficients, encouraging creative collision experiments. However, in Algodoo, momentum is primarily displayed in numerical form, and dynamic visualization requires manual activation. oPhysics [Walsh 2025] provides a low-cost online collision simulator, but its fixed scenarios and static numerical updates restrict deeper conceptual understanding. “Augmented Physics” [Gunturu et al. 2024] converts textbook diagrams into interactive animations, using multimodal techniques to extract content for physics simulation. However, a key limitation of these tools is the lack of intuitive, gesture-based control to directly initiate the physical simulations. In contrast to these tools, the proposed

PhysGest takes collisions and momentum conservation as its instructional focus. It not only supports the animation of textbook diagrams but also integrates gesture interaction and rigid-body simulation to generate realistic and continuous collision demonstrations, thereby deepening and broadening students' conceptual grasp of the law of momentum conservation.

2.3 Integration of Gesture Interaction and Physical Simulation

The integration of gesture interaction and physics simulation enhances the interactivity and intuitiveness of physics teaching by enabling natural gesture control combined with physically consistent simulations, thereby promoting deeper understanding of concepts such as momentum conservation. For instance, Johnson-Glenberg et al. [Johnson-Glenberg and Megowan-Romanowicz 2017] employed mixed reality and Kinect-based gesture capture to create immersive mechanics experiment environments, strengthening students' perception of dynamic physical processes. Solvang et al. [Solvang and Haglund 2022] integrated gesture control with a simulation platform, enabling students to explore friction and motion laws through bodily movements, thereby facilitating conceptual construction. Lu et al. [Lu et al. 2024] developed a gesture-interaction-based approach to manipulate molecular motion in virtual chemistry experiments, providing immersive teaching experiences and inspiring applications in physics education. Physink [Scott and Davis 2013] generates interactive physics animations from hand-drawn sketches, supporting dynamic simulation in physics education. PhysicsBook [Cheema and LaViola 2012] transforms hand-drawn physics diagrams into interactive animations, enhancing dynamic interactivity in physics teaching. Physica [Li et al. 2023] enables interactive physics simulations through a desktop robot, facilitating dynamic experimentation in physics education. However, these existing approaches are often constrained by predefined scenarios, offline training, or a lack of rigid-body simulation, limiting their flexibility in classroom teaching.

Building on these studies, our proposed PhysGest integrates image parsing, gesture interaction, and physics simulation to generate dynamic demonstrations consistent with the law of momentum conservation. With its low barrier to use and high interactivity, our PhysGest can assist instructor in explaining some core physics concepts and principles such as collision processes while enhancing students' understanding of conservation principles.

3 Method

As illustrated in Figure 2, the proposed PhysGest framework consists of three key modules: a) Diagram parsing module, b) Physical simulation module, and c) Video rendering module. These modules encompass semantic image understanding, gesture perception, physics-based simulation, and rendering synthesis, enabling a complete instructional workflow from interactive image-to-video generation. The proposed PhysGest applies to various physics teaching scenarios, including classroom demonstrations, concept instruction, and self-directed exploration.

3.1 Diagram Parsing Module

This module is designed to achieve a structured understanding of physical scenes in physics textbook diagrams while incorporating user gesture inputs to construct the initial motion states of objects, thereby providing physically consistent conditions for subsequent simulations. Specifically, large-scale pretrained models such as Grounded-SAM [Ren et al. 2024] are first employed for image segmentation. The segmented image is then passed to GPT-4V [Achiam et al. 2023], a multimodal large language model, for movability classification, dividing the image into foreground movable objects and static background regions. The physical properties of foreground objects are estimated, while edge information from the background is extracted to serve as boundary conditions for later physics simulations.

For gesture parsing, we employ MediaPipe Hands [Zhang et al. 2020], a high-fidelity, real-time hand tracking solution, to capture the positions of 21 hand landmarks. After capturing the continuous positional data of the 21 hand landmarks, we calculate the change in their positions between every adjacent frame, which allows us to quickly derive their spatial velocity and directional movement trends. Based on Newton's Second Law, these motion features are mapped into initial forces and torques acting on target objects, enabling dynamic coupling between user gestures and the physics simulation. In addition, to enhance visual consistency in video generation, a generative image inpainting model [Yu et al. 2023] is applied to restore occluded background regions, ensuring spatial completeness and temporal coherence in the subsequent video rendering model.

3.2 Physical Simulation Module

Using the physical properties of foreground objects extracted during the diagram parsing module and the initial force vectors applied to designated objects obtained from gesture parsing, the instance masks of foreground objects are fitted and converted into vector shape elements. Combined with boundary conditions extracted from the segmented background, a rigid-body physics simulation is performed in image space.

Our simulation approximates continuous motion by iteratively calculating each object's position, rotation, linear velocity, and angular velocity for each discrete time step. During the simulation, Newton's Second Law and the rigid body rotational equations are used to model the objects' motion under external forces, accurately computing trajectories, collisions, friction, and other physical phenomena to ensure consistency with Newtonian mechanics. For collision events, inter-object and object-boundary collisions are detected and resolved, with post-collision states calculated based on the law of momentum conservation, ensuring the physical consistency of the simulation.

3.3 Video Rendering Module

After the physical simulation is completed, the video rendering module converts the simulation results into high-quality dynamic videos. First, through image transformation and composition, the motion of foreground objects is applied to the input images and combined with the inpainted background to generate preliminary video frames. Next, the Lambertian [Pharr et al. 2023] shading model

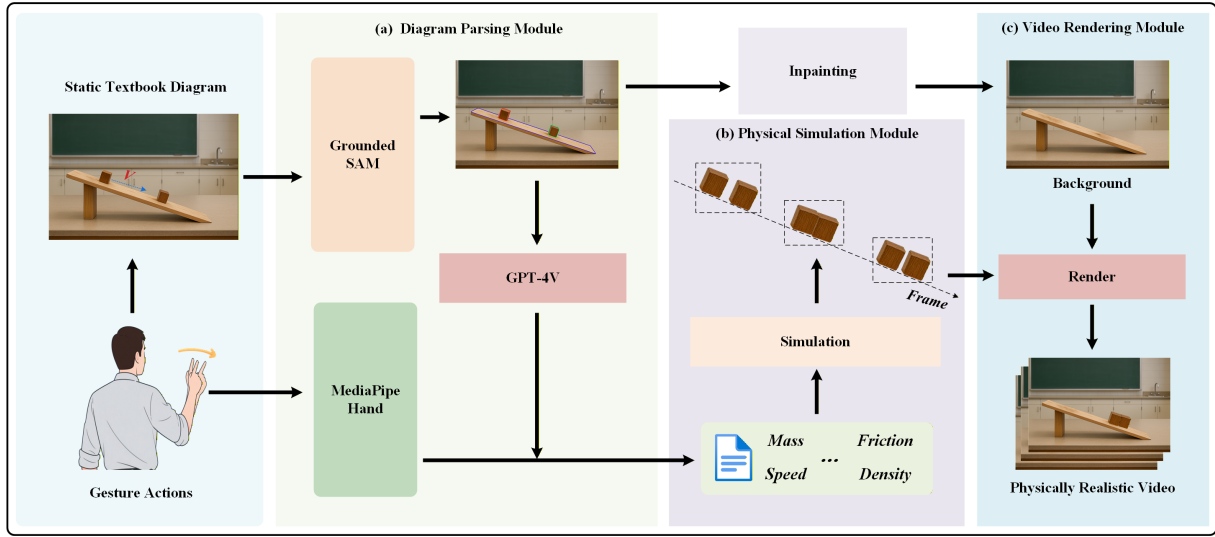


Figure 2: PhysGest consists of three modules: a) Diagram parsing module, b) Physical simulation module, and c) Video rendering module. The diagram parsing module is responsible for understanding the textbook diagram, extracting geometric and physical attributes from the static diagrams, and interpreting the instructor’s (or potentially student’s) gestures as physical forces. Subsequently, in the physical simulation module, a physics engine is used to simulate and calculate the objects’ motion trajectories. Finally, the video rendering module generates a physically realistic video by combining motion data with the inpainted background and applying video enhancement techniques to ensure both visual coherence and physical accuracy.

is used to relight each frame, simulating changes in illumination as the objects move, ensuring smooth and natural transitions in the video. Finally, a pre-trained video diffusion model [Chen et al. 2023] is employed to optimize the generated video, addressing issues such as missing background details and unnatural lighting effects. Through this series of processes, the resulting video not only adheres to physical laws but also achieves higher visual quality and realism, helping students better understand physics concepts intuitively.

4 Evaluation

4.1 Evaluation Design and Setup

We provide both quantitative and qualitative results to evaluate the effectiveness of the proposed PhysGest framework. First, we conducted a quantitative user study in which two groups of participants were asked to watch instructional videos on the concept of momentum conservation: one group viewed a traditional video combining textbook text and static diagrams, while the other group watched a video featuring dynamic demonstrations generated by PhysGest. Afterward, a questionnaire was administered to assess the overall learning experience and to examine the extent to which PhysGest enhanced participants’ understanding of momentum conservation. Second, qualitative results were presented by juxtaposing sample images from traditional text-based static diagrams and dynamic demonstration videos. The evaluation results indicate that dynamic demonstration videos generated by PhysGest achieve superior teaching effectiveness compared to traditional methods.

4.2 Quantitative Results

A total of 25 participants were recruited for the experiment (13 males, 12 females; mean age = 15 years, SD = 2.2) and were randomly assigned to either the experimental group or the control group. Before the learning task, all participants completed a short pre-test questionnaire to assess their prior knowledge of the concepts of collision and momentum conservation, ensuring that differences between groups were controlled.

The questionnaire covered three dimensions: self-efficacy [Bandura 1997], perceived enjoyment [Davis et al. 1992], and knowledge retention. The first two dimensions were measured using a 7-point Likert scale (1 = “strongly disagree”, 7 = “strongly agree”), with items adapted to the mechanics teaching context. Knowledge retention was assessed through test questions related to momentum conservation, e.g., “In which type of collision is momentum conserved but mechanical energy not conserved?” “Which physical quantity is necessarily conserved in all isolated collisions?” “If two objects collide elastically head-on on a smooth horizontal surface, which physical quantity is not necessarily conserved?” The accuracy rate (0%–100%, with an original maximum score of 10 points) was then normalized to a 0–7 scale for comparison.

The average questionnaire scores for the two groups on the self-efficacy dimension (questions SE_1 to SE_5) and the perceived enjoyment dimension (questions PE_1 to PE_5) are shown in Figure 3, while the knowledge retention test results are presented in Table 1. As shown in Table 2, participants in the experimental group, who used PhysGest-generated dynamic videos, achieved higher average scores in all three dimensions: self-efficacy (4.51), perceived enjoyment (4.47), and knowledge retention (3.65). To more rigorously

Table 1: Knowledge retention accuracy comparison (KR_1 to KR_4) between the control group (viewing the traditional video) and the experimental group (viewing the video generated by PhysGest).

	KR_1	KR_2	KR_3	KR_4
Control group	43%	44%	46%	44%
Experimental group	57%	59%	60%	58%

Table 2: Summary of average scores (0–7 normalized) across three dimensions, comparing the control group (viewing the traditional video) with the experimental group (viewing the video generated by PhysGest).

	Control group	Experimental group
Self-efficacy	3.92	4.51
Perceived enjoyment	3.88	4.47
Knowledge retention	3.18	3.65

assess the differences between groups, we conducted independent-samples t-tests on the data from both groups. The results, shown in Figure 4, indicate that the experimental group significantly outperformed the control group in all dimensions ($p < 0.05$). These findings demonstrate that PhysGest's physics-based dynamic visualizations have significant advantages and practical value in enhancing learners' understanding of collision mechanisms and conservation concepts, as well as in increasing learning motivation.

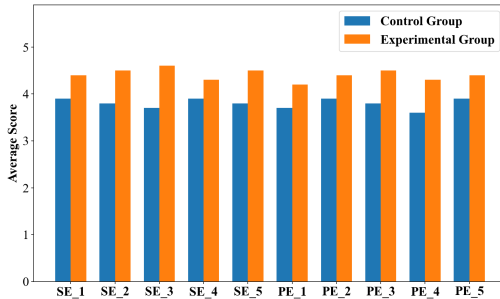


Figure 3: Visualizations of average scores for self-efficacy (question SE_1 to SE_5) and perceived enjoyment (question PE_1 to PE_5) comparing the control group (viewing the traditional video) with the experimental group (viewing the video generated by PhysGest).

4.3 Qualitative Results

In the qualitative demonstration, we used static textbook diagrams as input and compared them with the dynamic videos generated by PhysGest. Each sample consisted of three columns: the static diagram, a key frame before the collision, and a key frame after the collision (Figure 5). The results show that static diagrams can only present the initial state of the collision, making it difficult to

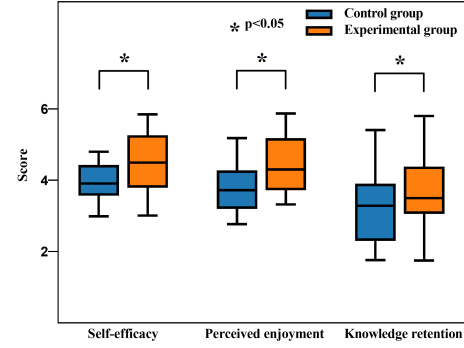


Figure 4: Significance analysis of scores for self-efficacy, perceived enjoyment, and knowledge retention in the control group (viewing the traditional video) and the experimental group (viewing the video by PhysGest)

fully understand the momentum conservation process. In contrast, PhysGest videos fully reconstruct the motion before the collision, the instant of impact, and the post-collision trajectory changes, strictly adhering to physical laws such as momentum conservation and avoiding errors such as sudden velocity changes or object penetration. This dynamic visualization transforms abstract theory into physically consistent and visually coherent animations, enabling learners to directly observe and analyze collision mechanisms, thereby bridging the cognitive gap between theoretical formulas and real-world dynamics, and enhancing both learning interest and depth of understanding.

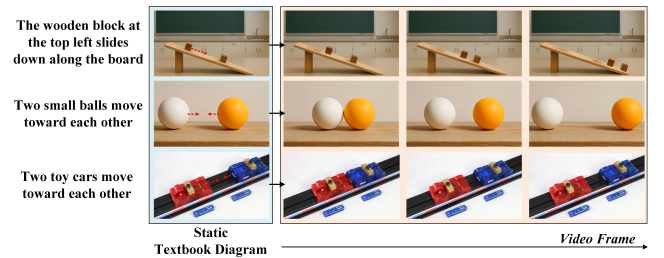


Figure 5: The qualitative results are presented through a comparison between the static diagrams from the textbook and the dynamic videos generated by PhysGest. The left side presents the original textbook diagram, while the right side displays corresponding key frames from the video generated by PhysGest.

5 Conclusion

In this paper, we present our PhysGest framework, which is closely aligned with the instructional objectives of physics curricula, particularly excelling in teaching physics concepts such as momentum conservation and collision mechanisms. Compared with traditional static diagram-based or conventional video-based teaching approaches, PhysGest transforms static textbook diagrams into

physics consistent dynamic demonstration videos, providing intuitive support for students' comprehension of these concepts. Consequently, PhysGest significantly enhances the understanding of abstract physics concepts by leveraging intuitive visual dynamics. Our experimental results show that PhysGest noticeably improves students' self-efficacy, perceived enjoyment, and knowledge retention.

Nevertheless, there are still aspects of PhysGest that need improvement. First, the current PhysGest framework is primarily optimized for two-dimensional collision scenarios and has limited capability in simulating three-dimensional or complex multi-body interactions. Future research could explore expanding PhysGest to three-dimensional scenarios in Virtual Reality and Augmented Reality. Second, the contextual richness of the generated videos could be further enhanced. For instance, integrating additional informational overlays, such as real-time force vectors, dynamic energy graphs, or other contextual visual cues, directly into the videos could provide learners with deeper insights into the underlying physical principles and foster stronger cognitive engagement throughout the learning process. Additionally, the current user study was conducted on a relatively small scale. Future research should involve larger sample sizes and longer teaching cycles to evaluate its sustained impact on long-term knowledge retention and the development of reasoning skills.

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